

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>  
2  int main(){  
3      int num;  
4      scanf("%d",&num);  
5      int count=0;  
6      count++;
```

```
7  ▼   for(int i=1;i<=num;++i){
8      count++;
9      count++;
10 ▼   if(num%i==0){
11      count++;
12
13      }
14  }
15  printf("%d",count);
16 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.